CELEBAL EXCELLENCE INTERNSHIP PROGRAM

DATA ENGINEERING

WEEK - 8

E-Commerce Analytics System

Project Outputs

SUBMITTED BY:-

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PROJECT OUTPUTS

1. Dataset Generation Output (generate_data.py)

```
PS E:\Celebal\Week8> python -u "e:\Celebal\Week8\scripts\generate_data.py"
customers.csv generated successfully.
products.csv generated successfully.
orders.csv generated successfully.
order_items.csv generated successfully.

All datasets generated successfully!
```

2. Data Cleaning Output (clean_data.py)

```
PS E:\Celebal\Week8> python -u "e:\Celebal\Week8\scripts\clean_data.py"
Cleaning completed successfully.
Issue report generated.
```

3. Data Quality Report (issues_report.txt)

```
data > issues_report.txt
Data Quality Report
====================

Invalid Emails: 10
CUST0020
CUST0056
CUST0103
CUST0114
CUST0196
CUST0241
CUST0419
CUST0440
CUST0447
CUST0471

Invalid Order References: 1
ORD99999
```

4. MySQL Database Schema (schema.sql)

- ecommerce_analytics
  - Tables
    - customers
    - order_items
    - orders
    - products
  - Views
  - Stored Procedures
  - Functions

- customers table schema

Field	Type	Null	Key	Default	Extra
customer_id	varchar(10)	NO	PRI	NULL	
customer_name	varchar(100)	YES		NULL	
email	varchar(100)	YES		NULL	
registration_date	date	YES		NULL	
customer_type	varchar(20)	YES		NULL	

- products table schema

Field	Type	Null	Key	Default	Extra
product_id	varchar(10)	NO	PRI	NULL	
product_name	varchar(100)	YES		NULL	
category	varchar(50)	YES		NULL	
subcategory	varchar(50)	YES		NULL	
cost_price	decimal(10,2)	YES		NULL	

- orders table schema

Field	Type	Null	Key	Default	Extra
order_id	varchar(10)	NO	PRI	NULL	
customer_id	varchar(10)	YES	MUL	NULL	
order_date	datetime	YES		NULL	
status	varchar(20)	YES		NULL	
region_code	varchar(20)	YES		NULL	

- order_items table schema

Field	Type	Null	Key	Default	Extra
item_id	varchar(10)	NO	PRI	NULL	
order_id	varchar(10)	YES	MUL	NULL	
product_id	varchar(10)	YES	MUL	NULL	
quantity	int	YES		NULL	
unit_price	decimal(10,2)	YES		NULL	
discount_percent	int	YES		NULL	

5. SQL Analytics Outputs (aggregations.sql)

- Revenue per Customer

customer_id	customer_name	total_revenue
CUST0229	Elizabeth Cabrera	829423.42
CUST0022	George Chapman	714814.43
CUST0197	James Gonzales	596272.09
CUST0150	Courtney Rodriguez	566942.24
CUST0124	Megan Le	546919.20
CUST0047	Robert Perez	536314.90
CUST0494	Emily Robinson	513092.55
CUST0415	William Graham	490644.71

- Revenue per Category

category	revenue
Home	12879780.89
Clothing	11276762.65
Electronics	10825948.26
Books	10456216.73

- Monthly Revenue

month	revenue
2024-07	1572303.11
2024-08	1770790.03
2024-09	1650254.39
2024-10	1992040.26
2024-11	2249616.54
2024-12	1672565.09
2025-01	927777.10
2025-02	980885.21
2025-03	1428966.34

- Top products by Quantity Sold

product_name	total_quantity
Lamp	264
Novel	248
T-Shirt	244
Tablet	242
Table	229
Mobile	207
Jacket	200
Sofa	193
Headphones	181
Chair	176

- Top Products by Revenue

product_name	revenue
Lamp	4393167.70
Novel	4181205.75
Tablet	3907591.10
T-Shirt	3692168.41
Table	3531725.35
Jacket	3446959.50
Shirt	3145129.26
Mobile	3059716.21
Sofa	2629295.29
Headphones	2581389.74

- Average Order Value (AOV) by Customer Type

customer_type	average_order_value
PREMIUM	106157.99
REGULAR	102105.09
VIP	123128.82

6. Advanced SQL Analysis Outputs (window_functions.sql)

- Customer Ranking by Lifetime Value using RANK() Window Function

customer_id	customer_name	lifetime_value	customer_rank
CUST0229	Elizabeth Cabrera	829423.42	1
CUST0022	George Chapman	714814.43	2
CUST0197	James Gonzales	596272.09	3
CUST0150	Courtney Rodriguez	566942.24	4
CUST0124	Megan Le	546919.20	5
CUST0047	Robert Perez	536314.90	6
CUST0494	Emily Robinson	513092.55	7
CUST0415	William Graham	490644.71	8
CUST0166	Sonia Day	488676.47	9
CUST0391	Rita Waller	481679.21	10
CUST0169	Rebecca Huang	453238.11	11

- Customer Ranking by Lifetime Value using DENSE_RANK() Window Function

Result Grid | Filter Rows: | Export: | Wrap Cell C

customer_id	customer_name	lifetime_value	dense_rank_customer
CUST0229	Elizabeth Cabrera	829423.42	1
CUST0022	George Chapman	714814.43	2
CUST0197	James Gonzales	596272.09	3
CUST0150	Courtney Rodriguez	566942.24	4
CUST0124	Megan Le	546919.20	5
CUST0047	Robert Perez	536314.90	6
CUST0494	Emily Robinson	513092.55	7
CUST0415	William Graham	490644.71	8
CUST0166	Sonia Day	488676.47	9
CUST0391	Rita Waller	481679.21	10
CUST0169	Rebecca Huang	453238.11	11

- Running Total Revenue Analysis using SUM() Window Function

order_day	daily_revenue	running_total
2024-07-20	400986.950000	400986.950000
2024-07-22	66997.530000	467984.480000
2024-07-23	660788.830000	1128773.310000
2024-07-24	1779.960000	1130553.270000
2024-07-27	402836.340000	1533389.610000
2024-07-31	38913.500000	1572303.110000
2024-08-02	537878.700000	2110181.810000
2024-08-05	87000.120000	2197181.930000
2024-08-06	121849.800000	2319031.730000
2024-08-12	112940.970000	2431972.700000

- Moving Average Revenue Analysis using AVG() Window Function

order_day	revenue	moving_average
2024-07-20	400986.950000	400986.95
2024-07-22	66997.530000	233992.24
2024-07-23	660788.830000	376257.77
2024-07-24	1779.960000	243188.77
2024-07-27	402836.340000	355135.04
2024-07-31	38913.500000	147843.27
2024-08-02	537878.700000	326542.85
2024-08-05	87000.120000	221264.11
2024-08-06	121849.800000	248909.54
2024-08-12	112940.970000	107263.63

- Monthly Revenue Analysis using Common Table Expression (CTE)

month	revenue
2024-07	1572303.110000
2024-08	1770790.030000
2024-09	1650254.390000
2024-10	1992040.260000
2024-11	2249616.540000
2024-12	1672565.090000
2025-01	927777.100000
2025-02	980885.210000
2025-03	1428966.340000

- Monthly Revenue Growth Analysis using CTE and LAG() Window Function

month	revenue	previous_month	growth_percent
2024-07	1572303.110000	NULL	NULL
2024-08	1770790.030000	1572303.110000	12.62
2024-09	1650254.390000	1770790.030000	-6.81
2024-10	1992040.260000	1650254.390000	20.71
2024-11	2249616.540000	1992040.260000	12.93
2024-12	1672565.090000	2249616.540000	-25.65
2025-01	927777.100000	1672565.090000	-44.53
2025-02	980885.210000	927777.100000	5.72
2025-03	1428966.340000	980885.210000	45.68
2025-04	1951904.610000	1428966.340000	36.60

- Customer Ranking by Lifetime Value using ROW_NUMBER() Window Function

Result Grid Filter Rows: Export:

customer_id	customer_name	lifetime_value	row_num
CUST0229	Elizabeth Cabrera	829423.42	1
CUST0022	George Chapman	714814.43	2
CUST0197	James Gonzales	596272.09	3
CUST0150	Courtney Rodriguez	566942.24	4
CUST0124	Megan Le	546919.20	5
CUST0047	Robert Perez	536314.90	6
CUST0494	Emily Robinson	513092.55	7
CUST0415	William Graham	490644.71	8
CUST0166	Sonia Day	488676.47	9

7. Customer Analytics Outputs (cohort_analysis.sql)

- Customer Cohort Analysis (First Purchase Month)

Result Grid Filter Rows:

customer_id	cohort_month
CUST0043	2024-07
CUST0209	2024-07
CUST0494	2024-07
CUST0048	2024-07
CUST0271	2024-07
CUST0147	2024-07
CUST0402	2024-07
CUST0197	2024-07

- Monthly Customer Retention Analysis

cohort_month	active_month	retained_customers
2024-07	2024-07	8
2024-07	2024-11	1
2024-07	2024-12	3
2024-07	2025-02	1
2024-07	2025-10	1
2024-07	2025-11	1
2024-07	2026-02	2
2024-07	2026-03	1

- One-Time Purchase Customer Analysis

customer_id	total_orders
CUST0004	1
CUST0006	1
CUST0013	1
CUST0015	1
CUST0016	1
CUST0017	1
CUST0018	1
CUST0021	1

- Repeat Customer Analysis

customer_id	total_orders
CUST0001	2
CUST0005	3
CUST0007	2
CUST0008	2
CUST0009	2
CUST0022	2
CUST0027	2
CUST0029	2

- Customer Purchase Frequency Segmentation

customer_id	total_orders	customer_segment
CUST0096	5	Occasional
CUST0032	4	Occasional
CUST0053	4	Occasional
CUST0124	4	Occasional
CUST0150	4	Occasional
CUST0197	4	Occasional
CUST0225	4	Occasional
CUST0294	4	Occasional
CUST0415	4	Occasional

- Customer Spend Tier Segmentation

customer_id	customer_name	total_spend	spend_tier
CUST0229	Elizabeth Cabrera	829423.42	High
CUST0022	George Chapman	714814.43	High
CUST0197	James Gonzales	596272.09	High
CUST0150	Courtney Rodriguez	566942.24	High
CUST0124	Megan Le	546919.20	High
CUST0047	Robert Perez	536314.90	High
CUST0494	Emily Robinson	513092.55	High

- RFM (Recency, Frequency, Monetary) Analysis

customer_id	customer_name	Recency	Frequency	Monetary
CUST0229	Elizabeth Cabrera	11	3	829423.42
CUST0022	George Chapman	339	2	714814.43
CUST0197	James Gonzales	161	4	596272.09
CUST0150	Courtney Rodriguez	264	4	566942.24
CUST0124	Megan Le	149	4	546919.20
CUST0047	Robert Perez	199	3	536314.90
CUST0494	Emily Robinson	272	3	513092.55
CUST0415	William Graham	[illegible]	4	[illegible]

8. **SQLite Reporting Tool (report_cli.py)**

```
PS E:\Celebal\Week8> python -u "e:\Celebal\Week8\scripts\report_cli.py"
E-Commerce Analytics Reporting Tool
Enter report type (daily/weekly/monthly): monthly
Enter start date (YYYY-MM-DD): 2024-02-01
Enter end date (YYYY-MM-DD): 2025-02-01

Report Result
------------------------------------------------------------
month | total_orders | revenue | unique_customers
------------------------------------------------------------
2024-07 | 8 | 2543790.0 | 8
2024-08 | 17 | 3155693.0 | 17
2024-09 | 20 | 3744833.0 | 20
2024-10 | 16 | 3723317.0 | 16
2024-11 | 20 | 4162354.0 | 20
2024-12 | 15 | 3613580.0 | 15
2025-01 | 14 | 2316035.0 | 14

Top 3 Products
------------------------------------------------------------
('Lamp', 85, 2524751.0)
('Table', 73, 2018532.0)
('Shirt', 46, 1868141.0)

Previous Period Comparison
------------------------------------------------------------
Previous period data not available
```

9. **Edge Case Testing (test_cases.py)**

```
PS E:\Celebal\Week8> python -u "e:\Celebal\Week8\scripts\test_cases.py"
Invalid order reference detected
Invalid discount detected
Zero quantity detected
Future order date detected
```